



Manipal School of Information Sciences

Cryptology Lab
LAB ASSIGNMENT 02

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Importance of Cryptography

HTTP and HTTPS Traffic Analysis Using Wireshark

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1 Problem Statement

Using Wireshark, investigate the transmission of dummy login credentials over HTTP and HTTPS in a controlled environment. Analyze the captured network traffic, compare how the credentials are transmitted in both protocols, and explain the role of encryption in protecting sensitive information.

The experiment is performed in a controlled environment using only dummy credentials and traffic generated by the experimenter.

1.1 Aim

To investigate and compare the transmission of dummy login credentials over HTTP and HTTPS using Wireshark, and to understand the importance of encryption in protecting sensitive information during network communication.

1.2 Objective

To capture HTTP and HTTPS network traffic using Wireshark and compare the visibility of dummy login credentials in both protocols. The experiment demonstrates how HTTP transmits application data without TLS protection, whereas HTTPS uses TLS encryption to protect the application data during transmission.

2 Experiment Procedure

The experiment is performed using a locally hosted test login application and dummy credentials.

1. Create a controlled local HTTP test server containing a dummy login form.
2. Start Wireshark and select the loopback network interface.
3. Open the HTTP login page in a web browser.
4. Capture the HTTP login request generated using dummy credentials.
5. Inspect the HTTP packet in Wireshark and observe the transmitted application data.
6. Set up a local HTTPS test server using TLS.
7. Open the HTTPS login page and enter the same dummy credentials.
8. Capture the HTTPS traffic using Wireshark.
9. Inspect the TLS packets and compare them with the HTTP packets.
10. Record the observations and explain how encryption protects sensitive information.

3 Experimental Setup

The experiment is performed in a controlled local environment. Wireshark is used as the packet analyzer, while a locally hosted test server is used to generate HTTP and HTTPS login traffic.

Component	Details
Operating System	Kali Linux
Packet Analyzer	Wireshark
Web Browser	Mozilla Firefox
HTTP Server	Local test server
HTTPS Server	Local TLS-enabled test server
Test Environment	Localhost / Controlled Environment
Credentials	Dummy username and password

Table 1: Experimental setup

4 HTTP Traffic Analysis

HTTP is an application-layer protocol that does not provide encryption for the application data transmitted between the client and server. Therefore, login information transmitted over HTTP can potentially be observed by inspecting the captured network traffic.

A local HTTP test server was used to generate dummy login traffic. The login request was captured using Wireshark and the packet contents were inspected.

4.1 HTTP Login Page

The local HTTP login page was opened in the browser and dummy credentials were entered.

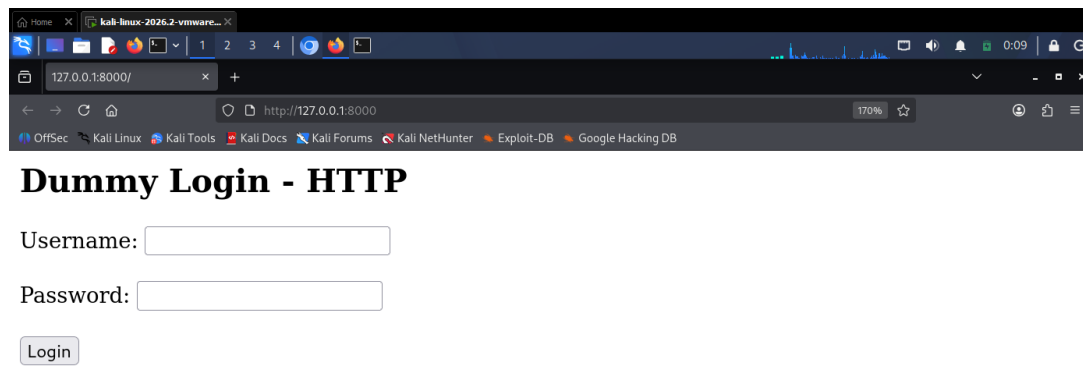


Figure 1: HTTP dummy login page

4.2 HTTP Login Request

The HTTP login request was captured using Wireshark. The captured traffic was examined to identify the HTTP request generated by the login operation.

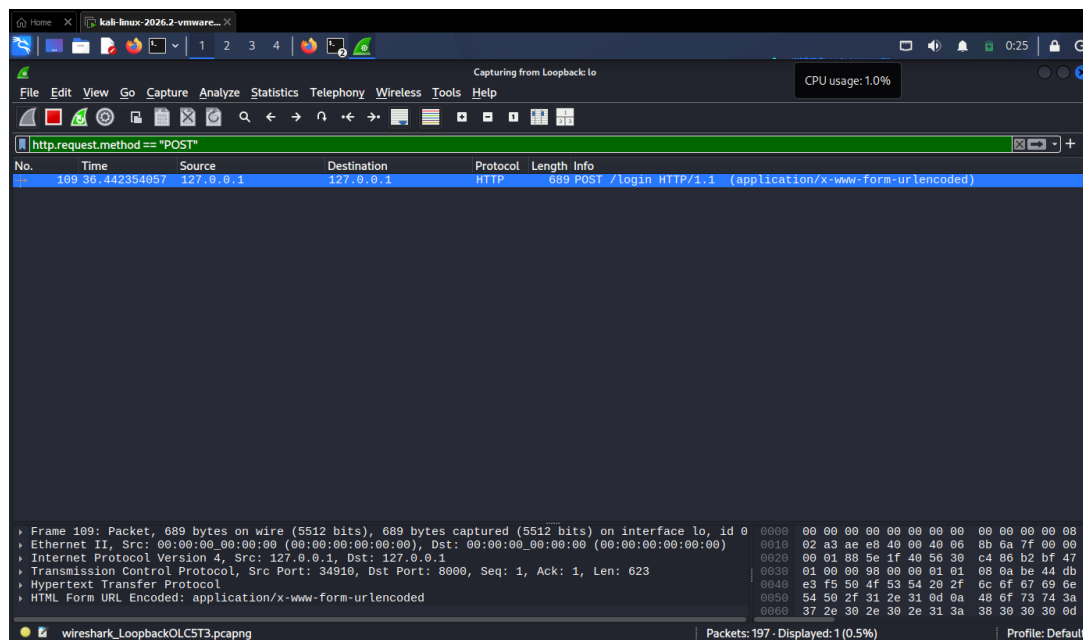


Figure 2: HTTP login request captured using Wireshark

4.3 HTTP Credential Visibility

The selected HTTP packet was inspected in Wireshark. Since HTTP does not encrypt the application data, the dummy login information can be visible in the transmitted

request data.

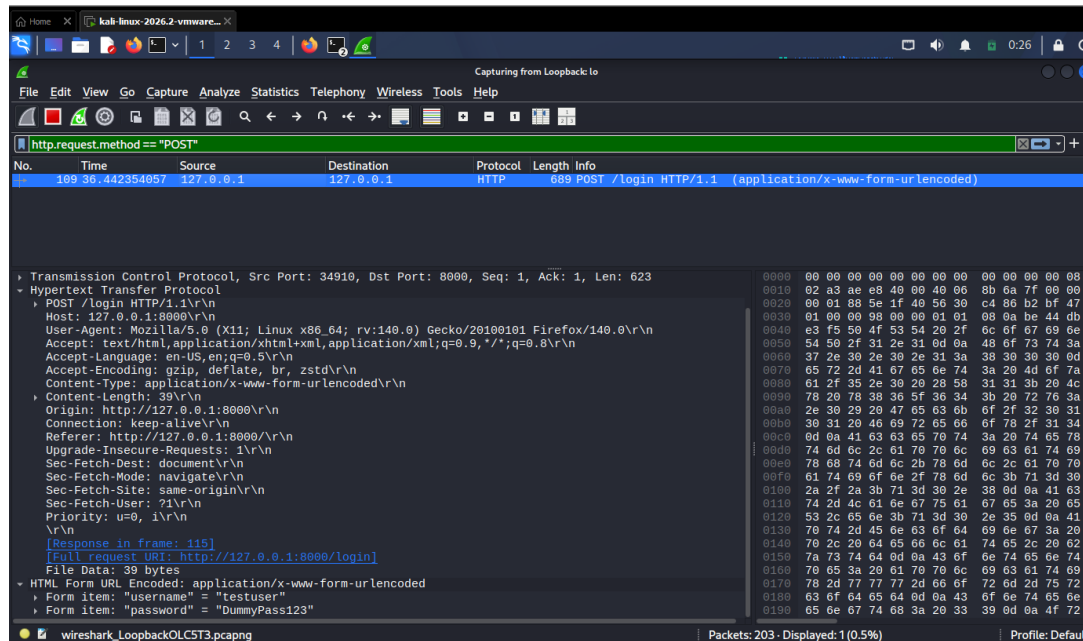


Figure 3: HTTP packet showing transmitted dummy credentials

5 HTTPS Traffic Analysis

HTTPS is HTTP transmitted over a TLS-protected connection. TLS provides encryption for application data exchanged between the client and server.

The same controlled environment was used to generate HTTPS login traffic using dummy credentials. The HTTPS packets were then captured and inspected using Wireshark.

5.1 HTTPS Login Page

The local HTTPS login page was opened in the browser. Dummy credentials were entered through the secure test connection.

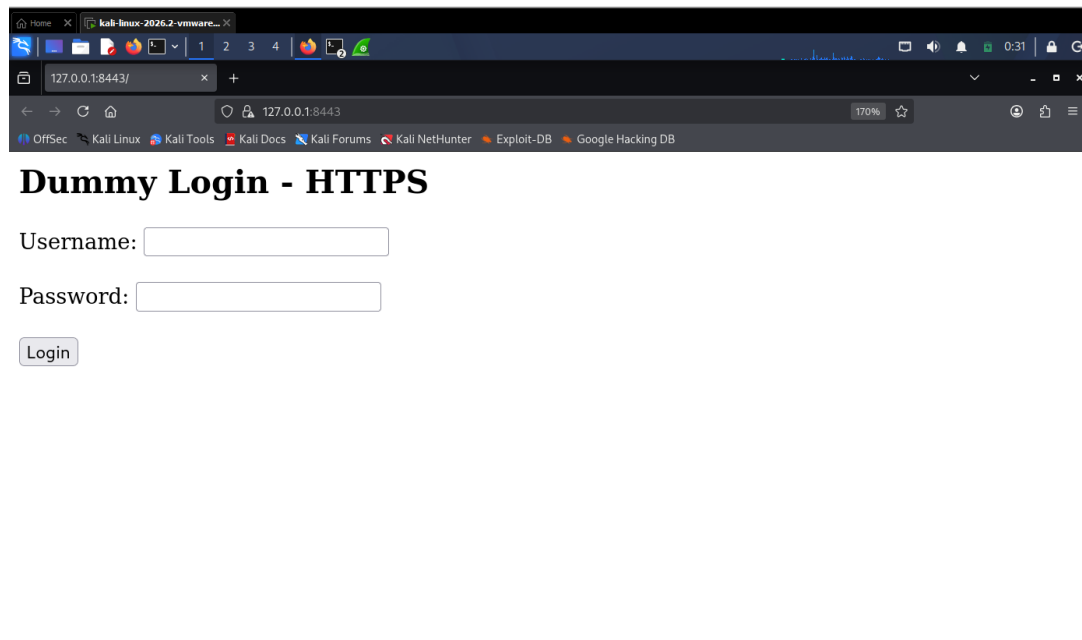


Figure 4: HTTPS dummy login page

5.2 HTTPS/TLS Traffic

The HTTPS login traffic was captured using Wireshark. TLS packets generated during the secure communication were observed in the packet capture.

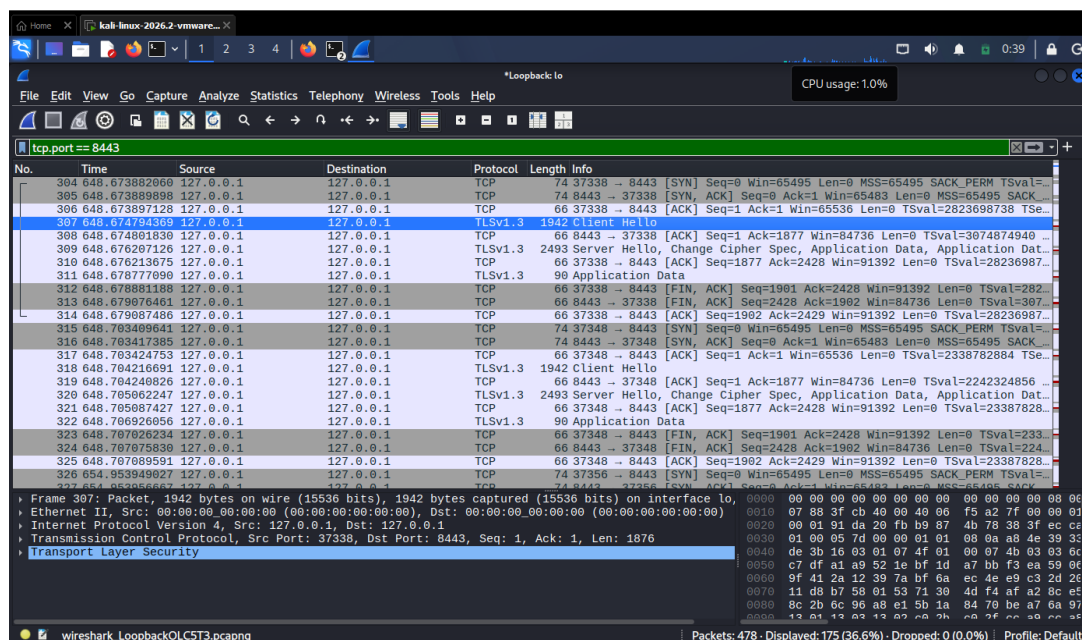


Figure 5: HTTPS/TLS traffic captured using Wireshark

5.3 HTTPS Encrypted Application Data

The TLS packet details were inspected to determine whether the dummy credentials were directly visible. Unlike the HTTP capture, the application data transmitted through the

HTTPS connection appears as TLS-protected data rather than readable HTTP form data.

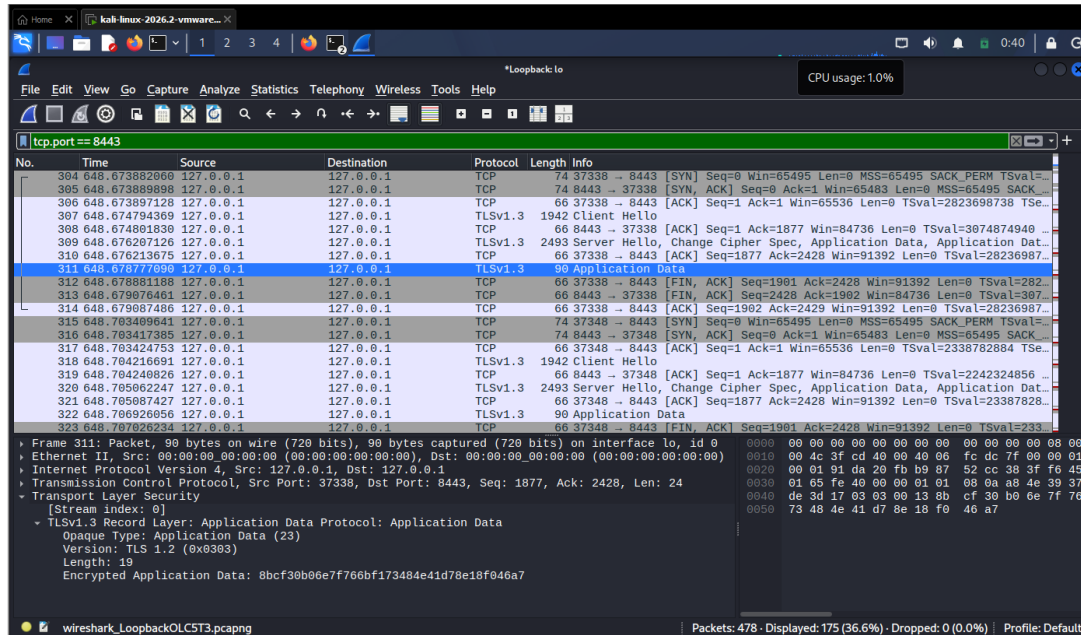


Figure 6: HTTPS packet showing TLS-protected application data

6 Comparison of HTTP and HTTPS

The captured HTTP and HTTPS traffic was compared based on the protection and visibility of the transmitted application data.

Feature	HTTP	HTTPS
Encryption	Application data is not protected by TLS encryption.	Application data is protected using TLS encryption.
Credential Visibility	Dummy credentials can be visible in the captured HTTP request.	Credentials are not directly visible as readable application data in the normal TLS packet capture.
Protocol	HTTP	HTTP over TLS
Confidentiality	Not provided by HTTP itself.	Provided through TLS encryption.
Packet Inspection	Application data can be inspected directly.	TLS-protected application data appears encrypted.

Table 2: Comparison between HTTP and HTTPS

7 Testing and Observations

The experiment was performed using dummy credentials in a controlled localhost environment.

Test Case	Experiment	Observation	Result
1	HTTP login	The HTTP login request was captured using Wireshark.	Pass
2	HTTP credential inspection	The dummy login data was visible in the HTTP request.	Pass
3	HTTPS login	The HTTPS login request generated TLS-protected traffic.	Pass
4	HTTPS packet inspection	The application data was not directly readable as HTTP form data.	Pass
5	Protocol comparison	HTTPS provided greater protection for the transmitted application data than HTTP.	Pass

Table 3: Testing and observations

8 Role of Encryption

Encryption converts readable plaintext into protected ciphertext so that unauthorized parties cannot directly understand the transmitted information.

In the HTTP experiment, the absence of TLS protection allowed the dummy login application data to be inspected in the captured request. This demonstrates the risk of transmitting sensitive information over an unencrypted connection.

In the HTTPS experiment, TLS was used to protect the application data transmitted between the client and server. The captured traffic showed TLS-protected communication instead of directly readable HTTP form data.

Therefore, encryption plays an important role in providing confidentiality for sensitive information such as login credentials and other private application data.

9 Conclusion

The experiment showed that HTTP transmits application data without encryption, making dummy credentials visible in captured traffic. In contrast, HTTPS uses TLS encryption to protect the transmitted data. Wireshark was used in a controlled localhost environment to capture and compare both protocols.

Thus, HTTPS provides better confidentiality and protection for sensitive information such as login credentials.